

G Deep 16

A fully synthesised deep house drum machine with a 16-step sequencer and a drum-bus chain.



Overview

G Deep 16 makes every sound from scratch - no samples. There are six voices: a kick, a snare, a clap, closed and open hi-hats and a rim shot. The kick is a sine wave whose pitch drops as it hits, with a click on the front and a drive that thickens the start. The snare and clap share a tonal body and a band of filtered noise. The hats are six square-wave oscillators tuned to the classic analogue metal ratios, blended with noise.

The voices then pass through a drum-bus chain: 1 COMPRESSOR, 2 3-BAND EQ, 3 DRIVE and 4 REVERB, then the MASTER volume. Each module has its own ON key.

A 16-step sequencer is built into the panel. It plays in time with REAPER's transport, has swing and a set of factory grooves, and can export its pattern as a MIDI item with the companion script. You can also play the voices from MIDI notes, or both at once.

Reach for it for

- Deep house, garage and Chicago-style grooves straight from the panel
- Synthesised kicks you can tune to the key of the track
- Wide house claps (WIDTH) and short, tuned rim shots
- Sketching a beat, then exporting it to MIDI to edit in REAPER
- Lo-fi or dubby drums through the drive and reverb modules

Playing it

G Deep 16 is an instrument: it has no audio input. It plays from its own sequencer, from MIDI notes, or both.

1. Insert G Deep 16 on a track.
2. To use the sequencer: switch PLAY on in the SEQUENCER section and start REAPER's transport. The steps follow the project's beat position (four beats to the bar), and the LCD shows the step that is playing and the tempo.
3. To play it from MIDI: arm the track and set its input to your MIDI keyboard, or draw notes in a MIDI item. Note 36 (C1) is the kick, 38 the snare, 40 the clap, 42 and 44 the closed hat, 46 the open hat and 37 the rim shot. Velocity sets the level (and on the kick, how long it rings).
4. Click a voice's name in the matrix to hear it on its own.

Triggering is sample-accurate. Incoming MIDI is passed on down the chain. CH in the MASTER section picks the MIDI channel it listens to (OMNI = all).

Controls

The header

KITS	Opens the list of factory kits. A kit sets every voice and module control; the sequencer pattern, the MIDI channel and the master volume are kept.
LCD	The top line shows the kit. The bottom line shows the sequencer: OFF, WAITING FOR PLAY (PLAY is on but the transport is stopped) or the step playing, with the project tempo.

S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

The voice matrix

One row per voice. On the left is the voice's name and MIDI note, in the middle its 16 steps, and on the right the knobs for that voice. Double-click any knob to reset it.

Voice name	Click it to hear the voice. It lights when the voice plays.
Steps	Click a step to go from off to hit to accent (shown with a white bar) and back to off. Hits play a little softer (72% for the kick, snare and clap, less for the hats and rim); accents play at full level. The steps are grouped in fours, and the playing step is highlighted.

KICK

PITCH	The kick's resting pitch, 30 - 120 Hz (48 Hz by default). Tune it to the key of the track.
DROP	How far above PITCH the hit starts, 0 - 600 Hz. More drop gives a punchier, more 'boomy' attack; none gives a plain sine thud.
P.DECAY	How fast the pitch falls back to PITCH, 2 - 200 ms. Short is a tight click-thump; long is a laser-like sweep.
DECAY	How long the kick rings, 50 - 2000 ms. Harder hits ring a little longer.
CLICK	The level of a short burst of bright noise at the very start, 0 - 1. It helps the kick cut through on small speakers.
DRIVE	Sub drive, 0 - 100%: soft saturation of the sine, strongest at the start of the hit. It thickens and fattens the kick.

LEVEL

The kick's level, -24 to +6 dB.

SNARE (body + level, snare and clap)

The snare and the clap are the same voice. The clap re-strikes its noise four times, 10 ms apart, before the tail (the classic clap), and keeps only a quarter of the tonal body.

TONE

The pitch of the tonal body, 100 - 400 Hz. It starts a little higher and drops quickly.

T.DECAY

How long the body rings, 10 - 500 ms.

TONE MIX

The balance between body and noise, 0 - 100% (40% by default). Low settings are a noisy, papery snare; high settings are a ringing, tom-like one.

LEVEL

The level of the snare and the clap, -24 to +6 dB.

CLAP (noise, snare and clap)**CUTOFF**

The centre of the band-pass filter on the noise, 500 Hz - 10 kHz. Lower is darker and thicker, higher is brighter and thinner.

RESO

The filter's resonance, Q 0.5 - 8. Higher settings give the noise a pitched, ringing quality.

N.DECAY

How long the noise lasts, 20 - 800 ms.

WIDTH

Delays the noise on the right side by 0 - 30 ms (12 ms by default) - the wide house clap. At 0 the noise is in the centre.

CLOSED HAT and OPEN HAT

Both hats share one metallic source and filter; each has its own decay. A closed hat cuts off a ringing open hat, as on the originals. The hats sit slightly to the right of centre.

TUNE	Retunes the six oscillators, -12 to +12 semitones.
TONE	The filter that shapes the hats, 3 - 16 kHz. Higher is thinner and more sizzling.
CLOSED	The closed hat's decay, 10 - 300 ms.
NOISE	Blends from the metallic oscillators (0%) to white noise (100%).
LEVEL	The level of both hats, -30 to +6 dB.
OPEN (open hat row)	The open hat's decay, 50 - 2000 ms.

RIM SHOT

Two short pitched pings and a click. It sits slightly to the left of centre.

PITCH	The pitch of the main ping, 600 - 3500 Hz; the second ping follows it lower down.
DECAY	How long it rings, 5 - 120 ms.
LEVEL	The rim's level, -30 to +6 dB.

SEQUENCER

PLAY	Switches the sequencer on. It plays only while REAPER's transport is running, locked to the project's beat position.
SWING	Delays every second 16th note, 0 - 75%. At 75% the off-beat 16ths arrive just over a third of a 16th late.
GROOVES	Opens the list of factory patterns. Choosing one replaces all six rows.
EXPORT MIDI	Sends the pattern to the companion export script (see below). The key lights briefly when pressed.
BARS (1 / 2 / 4 / 8)	How many bars the exported MIDI item lasts; the 16-step pattern is repeated to fill it.

Status line

Reminds you which script does the export, and shows 'export waiting' if you pressed EXPORT MIDI and the script is not running.

1 COMPRESSOR

A feed-forward peak compressor on the whole drum mix, the same gain on both sides.

ON	Switches the compressor in or out.
THRESH	The level where compression starts, -40 to 0 dB.
RATIO	1:1 to 20:1.
ATTACK	0.1 - 100 ms. Longer settings let the front of the kick through before it clamps down.
RELEASE	10 - 1000 ms. Shorter settings pump more.
MAKEUP	Gain after the compressor, -12 to +24 dB.
GR	The column of eight lights shows the gain reduction (about 1.5 dB per light).

2 3-BAND EQ

ON	Switches the EQ in or out.
LOW F / LOW	A low shelf: its frequency, 30 - 300 Hz, and gain, -15 to +15 dB.
MID F / MID Q / MID	A bell: its frequency, 150 Hz - 5 kHz, width (Q 0.3 - 6; higher is narrower) and gain, -15 to +15 dB.
HIGH F / HIGH	A high shelf: its frequency, 2 - 16 kHz, and gain, -15 to +15 dB.

3 DRIVE

ON	Switches the drive in or out.
TAPE / TUBE / SOFT	The saturation type. TAPE: smooth and symmetrical. TUBE: uneven, adding even harmonics for a warmer tone. SOFT: a gentle soft-clipper.
DRIVE	How hard the drums are pushed into the saturation, 0 - 24 dB. The level is compensated so it changes the colour more than the volume.
WET	The blend of driven and clean signal, 0 - 100%.

4 REVERB

ON	Switches the reverb in or out.
SIZE	The size of the room, 0 - 1.
DECAY	How long the tail lasts, 0 - 1.
DAMPING	How quickly the highs die away in the tail, 1 - 12 kHz. Lower is darker.
LOW-CUT	Cuts the lows going into the reverb, 50 - 1000 Hz (250 Hz by default), so the kick does not muddy the tail.
WET	The reverb level, -60 to 0 dB.

MASTER

VOLUME	The final output level, -24 to +12 dB.
CH	The MIDI channel G Deep 16 listens to: OMNI (all) or 1 - 16.
L / R meters	Output level, left and right. Click a meter to clear its peak hold.

Kits

Chosen from KITS in the header. A kit sets the voices and the four modules; the pattern, MIDI channel and master volume stay as they are.

Deep Classic	The default kit: a 48 Hz kick, all four modules on, light tape drive and a small room.
Dusty Warehouse	A lower, longer, more driven kick, a darker snare, tube drive and a bigger, darker reverb; duller hats with more noise.
Tight Tech	A short, clicky 52 Hz kick, a tight bright snare, fast 6:1 compression, soft-clip drive and a small dry room; short, bright hats.
Lo-Fi Dub	A deep 42 Hz kick with a long tail and heavy sub drive, a dark snare, hard tape drive and a dark reverb; dark hats with a long open hat.
Big Room Clap	A mostly-noise snare and clap with a long tail and 22 ms of width, a mid lift and a bigger room.
Soulful Warm	A long, round 46 Hz kick, gentle compression, tube drive and a warm reverb.
Minimal Click	A short, high, clicky kick, a bright snare, fast 8:1 compression, soft-clip drive and a tiny room; very short, bright hats.
Init (dry)	The default voices with all four modules switched off.

Grooves

Chosen from GROOVES in the SEQUENCER section. Each replaces the pattern in all six rows.

Deep Four	Four-on-the-floor kick, clap on 2 and 4, 8th-note closed hats and off-beat open hats.
Soulful	Four-on-the-floor with a pickup kick at the end of the bar, a ghost snare, clap on 2 and 4, busy hats and a rim.

Garage Shuffle	A skipping kick, ghost snares, clap on 2 and 4, a dense hat pattern and rim shots.
Broken Deep	A broken kick pattern with the snare on 2 and 4 and no clap; rims fill the gaps.
Classic Chicago	Four-on-the-floor, snare and clap together on 2 and 4 with a snare fill, 8th hats and off-beat open hats.
Dub Minimal	A sparse kick, one snare, soft claps and a scattered rim and hat pattern.
Off-beat Snare	Four-on-the-floor with the snare on every off-beat 8th, claps on 2 and 4, quarter-note closed hats and off-beat open hats.
Clear	Empties every row.

Exporting the pattern as MIDI

JSFX plug-ins cannot write files, so EXPORT MIDI hands the pattern to a ReaScript, "GFX G-Deep - export MIDI.lua", in the Scripts folder. The script makes a MIDI item at the edit cursor.

1. In REAPER, open Actions > Show action list, click New action > Load ReaScript, and pick "GFX G-Deep - export MIDI.lua".
2. Run the script. It keeps running in the background (its toolbar button lights, if you give it one) and exports every time you press EXPORT MIDI. Run it again to stop it.
3. Select the track that holds G Deep 16, put the edit cursor where the pattern should start, and choose BARS (1, 2, 4 or 8).
4. Press EXPORT MIDI. A MIDI item called "G-Deep pattern" appears at the edit cursor. If you pressed EXPORT MIDI before starting the script, it exports that pattern as soon as it starts.

The item uses MIDI channel 10 and the same notes as the plug-in: kick 36, snare 38, clap 40, closed hat 42, open hat 46, rim 37. Hits have velocity 100 and accents 127, the swing is kept, and the pattern is repeated for the number of bars chosen. You can then switch the sequencer off and play G Deep 16 from the MIDI item, editing the notes as you like.

Tips

- Tune the kick: set DROP to 0, find the note with PITCH, then bring DROP back for the punch.
- For a wider clap, raise WIDTH to 15 - 25 ms and keep the snare's TONE MIX low.
- Raise the reverb's LOW-CUT if the tail sounds boomy; the kick stays dry and tight underneath.
- Use accents sparingly - an accented hat every beat with plain hits in between gives the groove a natural lift.
- The sequencer pattern is saved with the project, so you can build a groove, save, and come back to it.

File: gfx-g-deep.jsfx